

Sanjeev Agrawal Global Educational University, Bhopal

School Name: School of Computer Technology			Department Name: Advanced Computing			
Recommended Programs: BTech (Hons) Computer Science & Engineering- AI/AIML/DS/DA/FSD/CSF					Semester: III	
Course Name	Operating System & Security		Course Code	CT25DCCS301T		
Credit Hours	L	T	P	N	Total Credits	3
	3	0	-	-		
Prerequisites (if any)	Computer System and Architecture					
Course Objectives	The course objectives are as follows: 1. This course should provide the students with a good understanding of Operating Systems including their architecture and all its components. 2. Good conceptions on all the subjects like processes, inter-process communication, semaphore, message passing, classical IPC problems, scheduling, memory management, file systems, security and protection mechanism, I/O hardware and software, deadlocks, etc. should be provided					
Course Content	Unit -1 (8) Introduction to OS -Architecture, Goals & Structures of O.S, Basic functions, Interaction of O. S. & hardware architecture, System calls, Batch, multiprogramming. Multitasking, time sharing, parallel, distributed & real -time O.S. Unit-2 (7) Process Management - Process Concept, Process states, Process control, Threads, Uni-processor Scheduling: Types of scheduling: Preemptive, Non preemptive, Scheduling algorithms: FCFS, SJF, RR, Priority, Thread Scheduling, Real Time Scheduling. System Calls like ps, fork, join, exec family, wait. Unit-3 (10) Concurrency control : Concurrency: Principles of Concurrency, Mutual Exclusion: S/W approaches, H/W Support, Semaphores, pipes, Message Passing, signals, Monitors, Classical Problems of Synchronization: Readers-Writers, Producer Consumer, and Dining Philosopher problem. Deadlock: Principles of deadlock, Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, System calls like signal, kill. Unit-4 (10) Memory Management Memory Management requirements, Memory partitioning: Fixed and Variable					

	Partitioning, Memory Allocation: Allocation Strategies (First Fit, Best Fit, and Worst Fit), Fragmentation, Swapping, and Paging. Segmentation, Demand paging, Virtual Memory: Concepts, management of VM, Page Replacement Policies (FIFO, LRU, Optimal, Other Strategies), Thrashing. I/O management & Disk scheduling: I/O Devices, Organization of I/O functions, Operating System Design issues, I/O Buffering, Disk Scheduling (FCFS, SCAN, C-SCAN, SSTF), RAID, Disk Cache. Inter Process Communication- Basic Concepts of Concurrency, Cooperating process, Advantage of Cooperating process, Bounded- Buffer - Shared-Memory Solution, Inter-process Communication (IPC), Basic Concepts of Inter-process Communication and Synchronization Unit-5 (10) OS & Security: OS Security Basics concepts: threats, vulnerabilities in OS, controls; risk; confidentiality, integrity, availability; security policies, security mechanisms; assurance; prevention, detection, deterrence, Malicious code: viruses, Trojan horses, worms Security in conventional operating systems: Memory, time, file, object protection requirements and techniques, Protection in contemporary operating systems, Identification and authentication, Identification goals, Authentication requirements, Human authentication, Machine authentication, Trusted OS, Advanced Operating System Basics of Network Operating System, Server Operating System and Real Time Operating System
Text Books	T1. A.Silberschatz & P.B. Galvin, 'Operating System concepts and principles', Wiley India, 8th ed., 2009. T2. A. Tanenbaum, 'Modern Operating Systems', Prentice Hall India, 2003.
References	R1. W.Stallings, 'Operating Systems: Internals and design Principles', Pearson Ed., LPE, 6th Ed., 2009. R2. M.J.Bach, 'Design of Unix Operating system', Prentice Hall, 1986.
Course Outcomes	At the end of this course students will be able to: 1. To learn and understand about OS and its Functionalities 2. To learn about process management, memory management, Device Management 3. Analyze computer networking and the security issues associated with computer networks 4. Demonstrate proficiency in configuring and operating various network security devices and operating systems 5. Analyze the essential concepts, algorithms, and protocols of applied operating system and network security

Mapping of Course outcome with Program Outcomes, PSO's, and Knowledge Levels (As per Blooms Taxonomy)

CO/ PO	P O1	P O2	P O3	P O4	P O5	P O6	P O7	P O8	P O9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3	Knowledge Levels (K1, K2, ..., K6)
CO 1	1													2	1	K2
CO 2		1		1	1								1	1		K2
CO 3	1		1	2	3								1		1	K4
CO 4	2		1	2	3								2	1	2	K3
CO 5	2	1		3	3								2	2	1	K4

High-3 Medium-2 Low-1

K1 =>Remember K2 =>Understand K3 =>Apply K4 =>Analyze K5 =>Evaluate K6 =>Create

Designed By: (Mr Nitin Choudhary)	Checked By: (Dr. Prashant Shrivastava)	Approved By: (Dr. Gourav Shrivastava)
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School Name: School of Computer Technology			Department Name: Advanced Computing			
Recommended Program: BTech (Hons) Computer Science & Engineering- AI/AIML/DS/DA/FSD/CSF					Semester: III	
Course Name	Operating System & Security Lab		Course Code	CT25DCCS301P		
Credit Hours	L	T	P	N	Total Credits	1
	-	0	2	-		
Prerequisites (if any)	Computer System and Architecture					
Course Objectives	Objective of this course is to: 1. Provide the students with a good understanding of the Operating System including its architecture and all its components. 2. Learn and understand OS and its functionalities. 3. Learn about process management, memory management, and device management. 4. Analyze computer networking and the security issues associated with computer networks. 5. Demonstrate proficiency in configuring and operating various network security devices and operating systems.					
Course Content	List of Experiments 1. Basic DOS, Linux commands and their Use. 2. Write a program to implement FCFS CPU scheduling algorithm. 3. Write a program to implement the SJF CPU scheduling algorithm. 4. Write a program to implement Priority CPU Scheduling algorithm. 5. Write a program to implement the Round Robin CPU scheduling algorithm. 6. Write a program to compare various CPU Scheduling Algorithms over different Scheduling Criteria. 7. Write a program to implement classical inter process communication problems (producer consumer). 8. Write a program to implement classical inter process communication problems (Reader Writers). 9. Write a program to implement classical inter process communication problems (Dining Philosophers). 10. Write a program to implement & compare various page replacement algorithms. 11. Write a program to implement & Compare various Disk & Drum scheduling Algorithms					

	12. Write a program to implement Banker's algorithms. 13. Write a program to implement Remote Procedure Call (RPC). 14. Write a Device Drivers for any Device or peripheral.
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References	R1. W. Stallings, 'Operating Systems: Internals and design Principles', Pearson Ed., LPE, 6th Ed., 2009. R2. M.J. Bach, 'Design of Unix Operating system', Prentice Hall, 1986.
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CO 2		1		1	1								1			K2
CO 3			1	2	3								1	1	1	K4
CO 4			1	2	3									2	2	K3
CO 5		1		3	3								2		3	K4

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Mr Chandan Kumar	Dr Prashant Shrivastava	Dr. Gourav Shrivastava
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